AVD-201 Dash Camera

User Manual

(Version 1.1)



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1. Product Introduction

AVD-201 integrates vehicle video monitoring, driving recorder, ADAS- advanced driver assistance system and DSM- driver status analysis system. It features the analog high-definition video recording, storage and playback. By using 3g/4g wireless transmission technology and GNSS technology, it provides real-time uploading of video recording, automobile driving record information, driving behavior analysis and alarm evidence.

1.1. Product characteristics and specifications

- Built-in high performance image processing chip
- H.264/H.265 encoding, high compression ratio
- Built-in front facing 1080P camera, Built-in rear facing 1080P camera, Support 1CH 1080P AHD
- DSM/ADAS

Power supply:

- 9-32V DC Wide Voltage Range
- Under-voltage, short and reversed plug-in protection
- Built-in smart power management system, shutdown under low voltage and low consumption when standby

Data storage:

- Special file management system to encrypt and protect the data
- Proprietary technology to detect the bad track of the SD card which can make sure the continuity of video and long service life of the SD card
- Built-in ultra-capacitor to avoid data loss and SD card damage caused by sudden power loss
- Support SD card storage, maximum 512G

Wireless module:

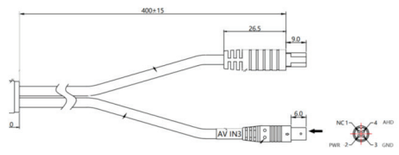
- Built-in 4G module
- Support GPS/BD/GLONASS, high sensitivity, fast positioning
- WiFi module (optional), frequency 2.4ghz
- Built-in 4G, GNSS and WIFI antenna

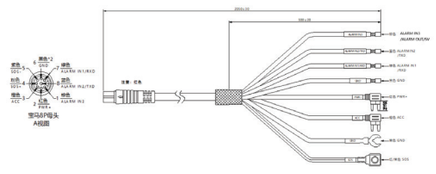
Product Technical Parameters:

Features	Parameters	Specification
System	Operating System	Embedded Linux
	Operating Langauge	English
Audio & Video	Compression Standard	H.265/H.264
	Image Resolution Ratio	1080P/720P/960H/D1/CIF
	Audio Compression	G.711A, G.711U, G.726
	Recording Mode	Synchronous recording of sound and video

Video Recording & Playback	Video recording mode	Automatic, Alarm
	Audio code rate	8Kb/s
	Video inquiry	It can be searched by channel & video type
Software Upgrade	Upgrade mode	Manual/Automatic/Remote
	Upgrade method	USB, SD Card, OTA
Interface	AV input	2CH Built-in 1080P camera 1CH 4pin BMW head port
	Ignition input	1
	SOS alarm	1
	Audio input	Built in Mic
	Audio output	Built-in 2W speaker
	SD card	1 SD card interface
	SIM interface	1 MICRO SIM interface
	USB interface	1 MICRO USB interface
	Status LED	PWR/RUN dual LED
Connectivity	GNSS	Built-in ceramic antenna, GPS+BD+GLONASS
	Wireless	4G
	WIFI	Frequency 2.4GHz
Other	Input Voltage	9V~32VDC
	Power consumption	Less than 10W
	Working temperature	-20°C - +70°C
	Storage capacity	1080P 600MB/ hour/channel H.265 1080P 1200MB/ hour/channel H.264
	Dimension	125.6*86.6*46.9mm

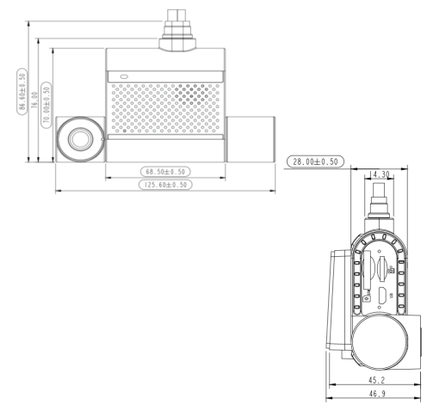
1.2. Wiring details





8pin type

1.3. Outline dimension drawing



2. Preparation before installation

2.1 Inspection of products and accessories

Before using this product, please check whether the accessories are complete. If there is any missing, please contact the support. The list of products and accessories is as follows:

Describe	Picture	Quantity	Describe	Picture	Quantity
Host machine		1 set	Accessories bag		1 set
Power cord		1 root	Mini DSM camera		1 (optional)

2.2 Preparation of tools required for installation

Please prepare auxiliary tools in advance before installation to facilitate site construction and installation. The details are as follows:

No.	Tool	Explain
1	Multimeter	Used to test whether there is electricity in the wire.
2	wire stripper	Stripping wires for use.
4	electrical tape	After wiring, the thread ends are wrapped and used.
5	cable ties	Line arrangement and wiring use
6	Extra Wires	When the standard line length of equipment is not enough, RVV conductor can be used to extend the connection, and the diameter of power extension line is not less than 1.0mm², and the signal extension line is not less than 0.5mm²

3. Device installation and construction

3.1. Host installation location

The Host device should be installed at the top of the front windshield. In order to avoid affecting the driver's sight, it is better to install it above the right co-driver seat and as close as possible to the center line of the vehicle (Right-hand drive cars are the opposite), as shown in the following figure:



The device has a built-in GNSS positioning antenna. In order to avoid interference, try to avoid the black shading area at the middle and upper part of the front windshield. The following figure shows the wrong installation method:



After the installation position is selected, paste the host device on the selected position. Before fixing, please wipe the windshield clean, which can be wiped with adhesive. When pasting, ensure the horizontal angle of the host device, and the left and right sides cannot be tilted. The dashcam needs to be pressed for more than 1 minute to ensure the bonding strength, and the wiring harness of the dashcam is hidden in the lining plate of the roof, as shown in the following figure:

3.2. Fitting installation

After opening the side cover of the device, insert the SIM card and memory (SD) card in the slots. Please pay attention to direction of SIM and SD card.



Note:
1. Please use Micro SIM card.
2. SD card: Please use SanDisk, Kingston or customized SD card (over 16G) purchased through regular channels.

After all accessories are installed, close the cover plate and fix it with screws (there are cover plates and fixing screws in the accessory bag). Please pay attention to the direction of the cover plate as in following figure.



After adjusting the camera angle, use the two screws on the back to secure the camera so it is not easily rotated.

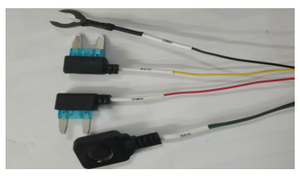


There are scales next to the camera to facilitate marking camera positions and batch adjustments.



3.3. Description of Power connection

Connect according to the marking on power wire. Before connecting, please check the power supply voltage, which should be within the range of 8-36V. The recommended working voltage is 12V or 24V.



Red wire: The red wire of the power supply is connected to the positive terminal of the main control power supply of the vehicle.
Black cable: The black wire of power supply should be connected to the GND of the vehicle power supply.
Yellow wire: The yellow wire of power supply (Ignition) should be connected to the ignition point of the vehicle.
Note:-
When connecting the power adapter to debug the device, please connect the red line and the orange line together to the positive pole of the power supply.
When the device is powered on, the LED will turn red, which is always on. When the device started working normally, the green LED will flash and the red LED will go out.

3.4. Wiring instructions

The wiring harness of the dashcam should be concealed, and it can be connected to the left A-pillar from the top inner liner, and then routed down the inner liner of the left A-pillar or the rubber strip, and then connected to the vehicle fuse box, as shown in the following figure:





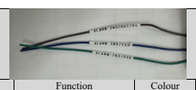
3.5. Main power search

Turn off the vehicle key, open the vehicle fuse box, use electroprobe or multimeter to measure and find it, unplug it after finding the fuse with electricity, and insert the red plug of the power cord into the plug with electricity in the fuse slot (if the fuse plug models are different, it is necessary to use adapter or broken wire connection, please connect it according to the actual situation).



3.6. I/O wire connection

The I/O line of the host computer is connected according to the requirements of the driving recorder, and the i/o line is marked accordingly. The definition is as follows:



Alarm	Function	Colour	Remark
Alarm input 1	Left turn indicator signal	Green	5~32v
Alarm input 2	Right turn indicator signal	Blue	5~32v
Alarm input 3	Output and input multiplexing	Brown	5~32v

Judgment method of left and right turn indicator signal lines: Turn on the left and right turn indicator signal lines and test them with Multimeter.
If other I/O wires are not used, the exposed wire ends must be cut off and wrapped with electrical tape to ensure that the exposed wire ends are not in contact with any part of the car body.

3.7. ADAS calibration

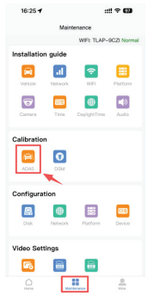
3.7.1. Vehicle Body Parameter Measurement

Measure the required vehicle parameters: car body width, the distance from the center of ADAS camera to the left side of the vehicle, the installation height of the camera (the height of the center of ADAS camera perpendicular to the ground) and the distance from ADAS camera to the front bumper of the vehicle, and record the test results. Later, enter the corresponding body parameters in the calibration of the "AtlantaVMS" APP "ADAS ".

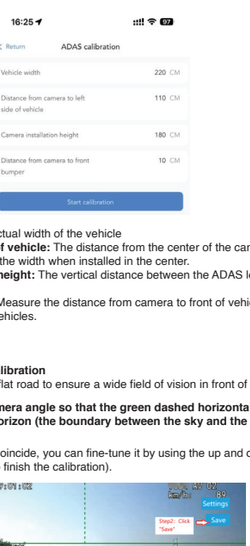
Note: Please connect alarm input 1 and alarm input 2 of the device to the vehicle's left indicator signal and right indicator signal, so that the lane departure alarm will not be triggered when the vehicle's indicator signal is activated.

3.7.2. Parameter setting before calibration

After connecting to the device's Wifi hot spot on your phone, open the AtlantaVMS app. After logging in and enter the **Maintenance interface**, click on the **"ADAS"** menu to enter the ADAS calibration interface.



Before calibration, you need to set the vehicle parameters first, and the parameters below must be filled in correctly. After filling the details, click on **"Start Calibration"**. The specific instructions are as follows:



Vehicle width: The actual width of the vehicle
Camera to left side of vehicle: The distance from the center of the camera to the left side, which is half the width when installed in the center.
Camera installation height: The vertical distance between the ADAS lens and the ground.
Camera to bumper: Measure the distance from camera to front of vehicle, which is generally 0 for large vehicles.

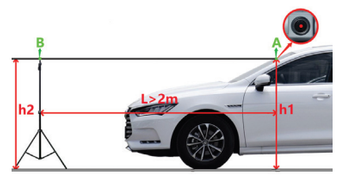
3.7.3. ADAS Calibration

Method1: Horizon calibration
Park the vehicle on a flat road to ensure a wide field of vision in front of it,

Adjust the ADAS camera angle so that the green dashed horizontal line coincides with the horizon (the boundary between the sky and the ground), then click "Save"
(if it can't completely coincide, you can fine-tune it by using the up and down arrow keys, and then click to finish the calibration).



Method2: Calibration with rod
Place a calibration rod at the position in front of the car (in front of the camera) (the distance L between points A and B must be more than 2m), and the height h2 of the calibration rod must be the same as the height h1 of the center point of the ADAS lens from the ground, as shown in the following figure:



Adjust the angle of the ADAS camera so that the top of the calibration rod coincides with the horizon position in the app (if it can't completely coincide, you can fine-tune it by using the up and down arrow keys), and then click Calibration Complete.



3.8. DSM Calibration

3.8.1. Installation of DSM camera (AV03 as DSM)

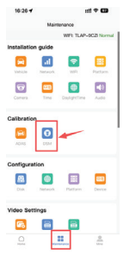
The DSM camera should be installed on the instrument panel at the right front of the steering wheel (Left-hand drive cars are the opposite). Tear off the 3m film on the base of the DSM camera, stick the camera on the instrument panel, and then fix it with self-tapping screws for the second time to prevent the position deviation of the equipment after long-term operation.



Please connect the DSM camera to the AV3 interface of the host computer(AVD-201), and then wrap the interface with insulating tape. If the video screen is black, please confirm whether the resolution of the camera and the device is the same.

3.8.2. Calibration of DSM camera (AV03 as DSM)

After connecting to the device's Wifi hot spot on your phone, open the AtlantaVMS app. After logging in and enter the Maintenance interface, click on the "DSM" menu to enter the DSM calibration interface.



Adjust the angle of the DSM camera so that the driver's face is in the center of the display interface. After fixing the angle, lock the three screws on the bracket to avoid angle deviation due to vibrations.



The pitch angle of DSM cannot exceed 30 °, otherwise it will affect the accuracy of the DSM algorithm.



3.8.3. Calibration of DSM camera (AV02 as DSM)

If you need use built-in channel 2 (AV02) as a DSM channel,



then please calibrate AV02 according to the following interface:



Step1: Choose le**ft-hand drive or right-hand drive** based on the driver's position,

Step2: Adjust the position of the DSM camera so that the driver's face is in the center of the red frame, and then click **"Save"**.

4. Device configuration using AtlantaVMS

The device can be connected to the backend server through the AtlantaVMS APP. Please download the latest AtlantaVMS APP first.

4.1. Download AtlantaVMS APP

Android APK: Search the "AtlantaVMS" in Google Play APP store, then install.
Or scan the QR code to download:
IOS APP: Search the "AtlantaVMS" in IOS APP store, then install.

4.2. AtlantaVMS APP connection

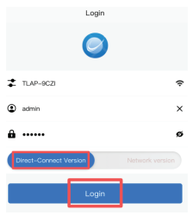
After the device is started, wait for about 1 minute, and use your mobile phone to search for WiFi hotspots. Connect your phone to the device's Wi-Fi hotspot. The hotspot name usually starts with **"TLAP-..."** and the password is **12345678**. After the connection is successful, proceed to the next setting operation.



Refer the serial number sticker of the device for Wi-Fi hotspot name

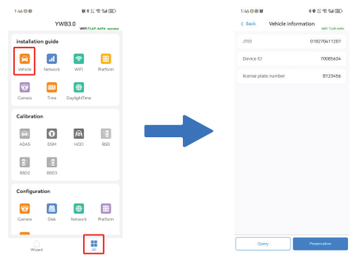
4.3. AtlantaVMS APP settings

Enter the AtlantaVMS login interface, keep the default username (admin) and password (123456), select **Direct-Connect Version** and click Login.



4.3.1. Vehicle information settings

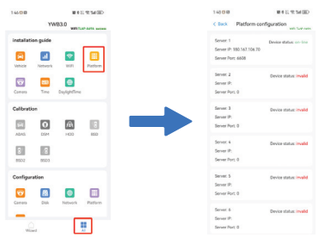
After logging in to the app, click on the configuration menu to configure device parameters. Select "Vehicle" in the left menu bar to set vehicle information. After the settings are completed, scroll down and click "Save" to save the parameters.



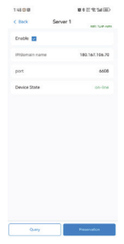
Don't change the JTID and Device ID. Only change the Plate number.

4.3.2. Platform information settings

Click on the "Platform" settings interface to set the server IP/domain name and port. This device supports 4 central server connections. Please make necessary settings.

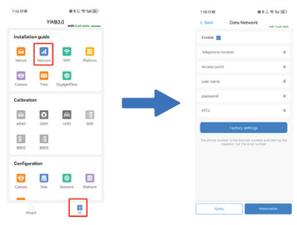


Select one of the columns to enter the IP address setting interface, as shown in the following figure. Please fill in the server address, port, and then click "Preservation" to save.



4.3.3. Network information settings

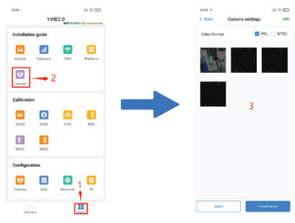
When using a APN private network, it is necessary to set the APN access point parameters in the "Network" configuration. The specific parameters are based on the parameters provided by the SIM card operator.



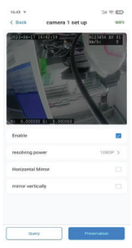
The specific parameters are subject to the parameters provided by the card vendor, and the corresponding apn or vpn values are filled in the "Access Point" setting item in the interface.
If this feature is not set correctly, the device will not be able to connect to 4G networks.

4.3.4. Camera mode setting

Click on the camera menu to enter camera mode settings.



This device can support connecting a third camera separately. Please set parameters based on the resolution of the camera. If the settings are incorrect, normal images will not be displayed. Please enable this channel first, then set the parameters and save them after completion



Enabled: all channels are enabled by default. If this channel is not used, uncheck it, otherwise a video loss alarm will be reported to the platform.
Resolution: Please set it according to the camera mode. If it is not set correctly, the image will not be displayed normally.

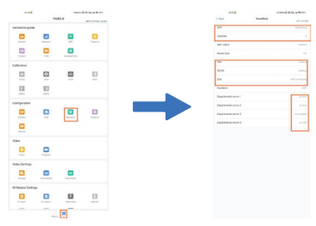
Horizontal Mirror: When checked, the device image will be horizontally inverted.
Vertical Mirror: When checked, the device image will be vertically inverted.

4.4. Product status view

In the "Configuration Status" column, you can view the status information of camera, disk, network, platform and equipment.

4.4.1. Network state

Select "all" in the lower right corner > "network" in configuration status in turn to view the "network" status, as shown below.



GPS: "Locked" is normal, and there are several other states-
1. Unauthorized means that the GPS function is not supported.
2. During positioning, the device is searching for GPS.

Satellite number: displays the number of satellites searched by the equipment and the satellite signal strength. When the number of satellites reaches more than 7 and the signal value reaches more than weak (strong, medium, weak and poor), the GSP data will be stable. If it cannot be reached, the installation location needs to be changed.

WIFI module: in the following states respectively.
1. Without authorization, the device does not support WIFI.
2. Yes, the device can search for available WIFI.
3. None, the device can't search for available WIFI.

WIFI status: check the networking status of wifi.
1. Not connected, not connected to wifi.
2. Connected, connected to wifi.

Module type: 3/4G module information, which uses LTE by default.
1. When no module is displayed, the device does not support the communication function or the module fails.

SIM card: the insertion detection status of SIM card.
1. Yes, the device has detected the SIM card.
2. No, the device has not detected the SIM card.
3. Abnormal. The device detects the SIM card, but it cannot read the SIM card correctly.

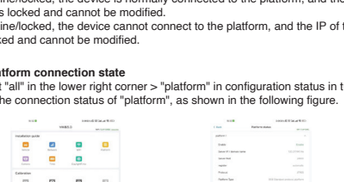
3G/4G signal: It indicates that it is not necessary to check whether the antenna is plugged in or whether the APN is set correctly at all times. Generally, it indicates that the signal strength is weak-strong, which means normal.

Dialing: Dialing networking status of SIM card.
1. Connection. The SIM card of the device is connected normally.
2. No connection. The device is not connected to the SIM card network.
3. During dialing, the SIM card of the device is dialing the Internet.
4. If the registration fails, check whether the SIM card is open or in arrears, and whether the APN is set correctly.

Network type: generally divided into WIFI(WIFI connected to the Internet) and 2/3/4G(SIM card connected to the Internet). When both networks can access the Internet, the priority is WiFi > 2/3/4G.

1~4: 808 protocol platform connection status of the server (this status is not displayed if the device does not support 808 protocol).
1. Online is connected to the platform normally, while offline is not connected to the platform.
2. If it is not enabled, the IP enabled status of this road is not checked. The authorization expires, please contact the relevant business.
3. Conflict, which is the same as the IP set by other routes, and there is conflict.
4. Online/locked, the device is normally connected to the platform, and the IP of this road is locked and cannot be modified.
5. Offline/locked, the device cannot connect to the platform, and the IP of this road is locked and cannot be modified.

A. Platform connection state
Select "all" in the lower right corner > "platform" in configuration status in turn to view the connection status of "platform", as shown in the following figure.



In the "Platform" information, you can view the setting information and connection status of the four platforms. When the "Equipment Status" shows "Online", it means that the equipment and the modified platform are connected normally.

4.4.2. Device status

Select "all" in the lower right corner > "equipment" in configuration status, and you can view the information of "equipment", as shown in the following figure.

In the "Device" state, you can check the software and hardware version of the device, the version of the 4G\GPS module, etc. You can look at the basic information first when troubleshooting the problem.

4.4.3. Disk state

Select "all" in the lower right corner > "Disks" in the configuration status to view the information of "Disks", as shown in the following figure.

You can see the status of SD1 in the "Disk" information.
Name: SD1 corresponds to device TF1. If only one disk information is displayed, the device only supports a single card.
Total capacity: the total capacity of the current TF card
Remaining capacity: the remaining capacity of the current TF card, which is 0 when the video is overwritten.

5. Frequently asked questions

Phenomenon	Phenomenon analysis	Processing method
Unable to boot	The power supply is not connected correctly.	Connect the wiring as required and ensure that the input voltage is within 8-36V.
	Power cord fuse	Eliminate the cause of burning and replace the fuse.
Can't connect to the center	Parameter setting error (server not connected)	Reset according to the instructions
	SIM card arrears (dialing or registration failure)	Recharge after inquiry
	APN parameter error (dialing or registration failure)	Check the parameters with the operator and reset them.
	Is the SIM contact good (without SIM card)	Reinsert and install SIM card.
No video recording	Unformatted disk	Format disk locally or remotely.
	Disk damage	Replace disk
	The disk lock is unlocked.	Lock the disk lock, and there is a prompt on the disk status in the upper right corner.
Unable to locate	Vehicles are in underground parking lots and tunnels.	Leave the area
No image display	Incorrect interface definition	Check whether the interface definitions are consistent.
	Mode setting error	Follow the camera mode for setting.
	Camera damage	Replace the camera with a new one

After the installation is completed, you need to check the following:

Check item	Has the camera protective film been torn off?	
	Can the alarm be generated normally?	
	Is 4G dialing successful?	
	Did the device successfully connect to the server?	
	Does the platform check whether the equipment is positioned?	
	Is the platform alarm data normal?	
	Is there any training for customers? Equipment installation Equipment debugging Equipment alarm test Platform use method Frequently asked questions and answers	